Galois Theory for Programmers

From Symmetries to Unsolvability in Lean 4

A self-contained journey through abstract algebra,
from group axioms to the insolvability of the quintic,
with every definition formalized and every theorem proved.

Draft — March 16, 2026

Contents

Preface

This book tells the most dramatic story in the history of mathematics, and asks you to formalize it in a proof assistant.

The story begins in the 1770s, when Joseph-Louis Lagrange noticed that the roots of a polynomial equation have *symmetries*—you can rearrange them in certain ways without breaking the algebraic relationships between them. Over the next sixty years, a succession of mathematicians—Gauss, Abel, Galois, Cauchy—gradually uncovered a deep connection between these symmetries and the solvability of the equation. The climax came in 1832, when Évariste Galois, a twenty-year-old political radical in Paris, wrote down the definitive answer the night before he was killed in a duel: a polynomial equation can be solved by radicals if and only if its *group of symmetries* has a certain structural property called *solvability*.

This is not just a historical curiosity. The mathematical framework Galois created—now called Galois theory—turns out to be fundamental to modern cryptography (the AES encryption algorithm does arithmetic in a Galois field), coding theory (Reed-Solomon codes work over finite field extensions), and the deepest parts of number theory and algebraic geometry.

What this book is. A self-contained, zero-to-hero treatment of Galois theory for programmers, with all definitions, theorems, and proofs formalized in Lean 4. No prior knowledge of abstract algebra is assumed. If you can write a function and follow a logical argument, you have everything you need. Every concept is accompanied by:

- A *historical narrative* explaining where the idea came from and who discovered it.
- A *formal definition* in standard mathematical notation.
- A *Lean 4 formalization*: a typeclass, structure, or theorem statement that you can type-check.
- *Worked examples* computed by hand and verified in Lean.
- *Exercises* ranging from "prove this basic property" to "formalize this entire section."

What this book is not. A graduate algebra textbook. We sacrifice generality for clarity: we work with finite groups and finite-degree field extensions, we avoid category theory (though we point out connections), and we prefer explicit computation over abstract elegance when the two conflict. The goal is understanding, not maximal generality.

Prerequisites. Comfort with Lean 4 syntax: defining inductive types, writing functions, using tactics (`intro`, `rfl`, `simp`, `omega`, `cases`, `induction`). No prior abstract algebra.

Lean 4 setup. All code in this book is written for vanilla Lean 4 (no Mathlib dependency). We build every algebraic structure from scratch so you see exactly what's happening inside the typeclasses.

Part I

The Algebraic Toolkit

Chapter 1

Groups — The Algebra of Symmetry

> "The art of doing mathematics consists in finding that special case which contains all the germs of generality."
>
> *David Hilbert*

1.1 Where Groups Come From

In the 1770s, Joseph-Louis Lagrange was studying a question that had occupied mathematicians for centuries: *how do you solve polynomial equations?*

The quadratic formula was ancient knowledge. The cubic and quartic formulas had been discovered in the 1500s by the Italian mathematicians del Ferro, Tartaglia, Cardano, and Ferrari—amid a web of secrecy, betrayal, and public mathematical duels that reads more like a Renaissance thriller than a mathematics paper. But for degree five and higher, nobody could find a formula.

The Italian Algebraists

Scipione del Ferro secretly discovered the cubic formula around 1515 but told only his student Antonio Fior. When Niccolò Tartaglia independently found the formula in 1535, Fior challenged him to a public contest. Tartaglia won decisively, then shared his secret with Gerolamo Cardano under a sworn oath of secrecy. Cardano later published it anyway (in his 1545 *Ars Magna*), claiming del Ferro's priority justified breaking the oath. Tartaglia was furious. The quartic formula was found by Cardano's student Ludovico Ferrari, who then challenged Tartaglia to another public contest—and won.

This entire soap opera happened because a general formula for solving equations was considered the ultimate mathematical achievement. The quest for the quintic formula would consume mathematicians for the next 250 years.

Lagrange's approach was novel: instead of trying to find the quintic formula directly, he asked *why* the existing formulas worked. He noticed that the key step in solving a cubic or quartic is finding an expression in the roots that takes on fewer values when the roots are *permuted*. For example, if a cubic has roots r_1, r_2, r_3, the expression $r_1 + \omega r_2 + \omega^2 r_3$ (where $\omega = e^{2\pi i/3}$) takes only three values under all six permutations of the roots. This "reduction in the number of values" is what makes the equation solvable.

Lagrange didn't have the word "group"—that term wouldn't be coined until Galois used it in 1832—but he was seeing *symmetry*: the permutations of the roots, and the structure of how those permutations compose with each other, control whether the equation can be solved.

Intuition

Think of a group as a mathematical way to talk about *symmetry*. The symmetries of any object—a geometric shape, a set of roots, a crystal structure—form a group. The group captures *which rearrangements are possible* and *how they compose*.

1.2 The Group Axioms

After Lagrange, it took nearly a century for the abstract concept of a "group" to crystallize. Cauchy (1815) studied permutation groups systematically. Cayley (1854) gave the first abstract definition. But it was only in the late 1800s, through the work of Kronecker, Weber, and Frobenius, that the modern axioms became standard.

Definition 1.1 — Group

A **group** is a set G together with a binary operation $\cdot : G \times G \to G$ satisfying:

(i) **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.

(ii) **Identity:** There exists an element $e \in G$ such that $e \cdot a = a \cdot e = a$ for all $a \in G$.

(iii) **Inverses:** For each $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

If additionally $a \cdot b = b \cdot a$ for all $a, b \in G$, the group is called **abelian** (or **commutative**).

Why "abelian"?

Named after Niels Henrik Abel (1802–1829), the Norwegian mathematician who proved the general quintic is unsolvable by radicals. We will meet Abel's theorem in Chapter 10. He died of tuberculosis at 26, in poverty, two days before a letter arrived offering him a prestigious professorship in Berlin.

1.2.1 Lean 4 Formalization

We define groups as a typeclass, just as Lean's standard library defines `Add` or `Mul`:

Lean 4

```lean
class Group (G : Type) where
  mul : G → G → G
  one : G
  inv : G → G
  mul_assoc : ∀ a b c : G, mul (mul a b) c = mul a (mul b c)
  one_mul   : ∀ a : G, mul one a = a
  mul_one   : ∀ a : G, mul a one = a
  inv_mul   : ∀ a : G, mul (inv a) a = one
  mul_inv   : ∀ a : G, mul a (inv a) = one

-- Notation
instance [Group G] : Mul G where mul := Group.mul
instance [Group G] : One G where one := Group.one
instance [Group G] : Inv G where inv := Group.inv
```

Intuition

Notice the parallel with programming interfaces. A `Group G` instance is exactly an *implementation* of the group interface for the type `G`. The axioms (`mul_assoc`, `one_mul`, etc.) are *contracts* that any implementation must satisfy—and in Lean, satisfying a contract means providing a *proof*.

1.3 First Examples

1.3.1 The Integers Under Addition

The simplest infinite group: $(\mathbb{Z}, +)$ with identity 0 and inverse $-n$.

Lean 4

```
instance : Group Int where
  mul := (· + ·)
  one := 0
  inv := (- ·)
  mul_assoc := Int.add_assoc
  one_mul   := Int.zero_add
  mul_one   := Int.add_zero
  inv_mul   := Int.neg_add_cancel
  mul_inv   := Int.add_neg_cancel
```

Every axiom is discharged by a single lemma from Lean's standard library. This is because the integers were *designed* with these properties in mind—the group axioms are capturing something that was always there.

1.3.2 Modular Arithmetic — $\mathbb{Z}/n\mathbb{Z}$

Gauss and the *Disquisitiones*

Carl Friedrich Gauss published his *Disquisitiones Arithmeticae* in 1801, at the age of 24. It systematized the theory of modular arithmetic ("clock arithmetic") and introduced the notation $a \equiv b \pmod{n}$. The *Disquisitiones* is one of the most influential mathematics books ever written—it transformed number theory from a collection of scattered results into a coherent discipline. Gauss didn't use the word "group," but the structures he studied—integers modulo n, primitive roots, quadratic residues—are all group-theoretic.

The group $\mathbb{Z}/n\mathbb{Z}$ consists of the integers $\{0, 1, \ldots, n-1\}$ with addition modulo n. In Lean, this is naturally `Fin n`:

Lean 4

```
-- We represent ℤ/nℤ as Fin n (for n ≥ 1)
-- Fin n = { val : Nat // val < n }
-- Addition is modular: (a + b) mod n

def Fin.addMod (a b : Fin n) : Fin n :=
  ⟨(a.val + b.val) % n, Nat.mod_lt _ (by omega)⟩

def Fin.negMod (a : Fin n) (h : n > 0) : Fin n :=
  ⟨(n - a.val) % n, Nat.mod_lt _ h⟩
```

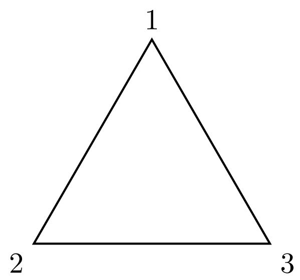
Exercise 1.1

Prove that `Fin n` with `addMod` satisfies the group axioms for any $n \geq 1$. The hardest part is associativity—you will need properties of the modular arithmetic operations. *Hint:* use `Nat.add_mod` and `omega`.

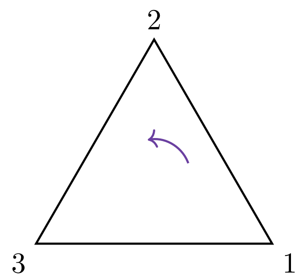
1.3.3 Symmetries of a Triangle — S_3

The **symmetric group** S_n is the group of all permutations of n elements. S_3—the symmetries of $\{1, 2, 3\}$—is the smallest non-abelian group (6 elements, and the order of multiplication matters).

Think of it as the symmetries of an equilateral triangle: three rotations (by 0°, 120°, 240°) and three reflections.



e (identity)



r (rotate 120°)

Lean 4

```
-- S₃ as an explicit enumeration
inductive S3 where
  | e                    -- identity
  | r                    -- rotation 120°
  | r2                   -- rotation 240°
  | s                    -- reflection (fixes vertex 1)
  | sr                   -- reflection (fixes vertex 3)
  | sr2                  -- reflection (fixes vertex 2)
  deriving DecidableEq, Repr

namespace S3

-- Multiplication table (composition of symmetries)
def mul : S3 → S3 → S3
  | e,   x   => x
  | x,   e   => x
  | r,   r   => r2
  | r,   r2  => e
  | r2,  r   => e
  | r2,  r2  => r
  | s,   s   => e
  | s,   r   => sr2
  | s,   r2  => sr
  | r,   s   => sr
  | r2,  s   => sr2
  | sr,  sr  => e
  | sr2, sr2 => e
  | s,   sr  => r
  | s,   sr2 => r2
  | sr,  s   => r2
  | sr2, s   => r
  | r,   sr  => s
  | r,   sr2 => sr
  | r2,  sr  => sr2
  | r2,  sr2 => s
  | sr,  r   => s
  | sr,  r2  => sr2
  | sr2, r   => sr
  | sr2, r2  => s
  | sr,  sr2 => r2
  | sr2, sr  => r

def inv : S3 → S3
  | e   => e
  | r   => r2
  | r2  => r
  | s   => s
  | sr  => sr
  | sr2 => sr2

end S3
```

Exercise 1.2

Prove that `S3` with the multiplication table above satisfies the group axioms. Since `S3` is finite with 6 elements, associativity can be proved by exhaustive case analysis: `decide` or pattern matching on all $6^3 = 216$ triples. *Warning:* this is tedious but instructive. Consider using `native_decide` for efficiency.

Exercise 1.3

Prove that S_3 is not abelian by exhibiting a concrete counterexample: find $a, b \in S_3$ such that $a \cdot b \neq b \cdot a$, and prove the inequality in Lean.

1.4 Euler and Modular Exponentiation

Euler's Theorem

Leonhard Euler (1707–1783) is the most prolific mathematician in history, with over 800 published papers. Among his countless contributions, his work on modular arithmetic laid the foundations for what we now recognize as group theory. Euler's theorem—$a^{\varphi(n)} \equiv 1 \pmod{n}$ for $\gcd(a, n) = 1$—is, in modern language, a consequence of Lagrange's theorem applied to the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$. Euler proved it by direct combinatorial reasoning in 1736, more than a century before the group-theoretic framework existed to express it cleanly.

Euler's totient function $\varphi(n)$ counts the integers from 1 to n that are coprime to n. In group-theoretic terms, $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$—the order of the multiplicative group modulo n.

Lean 4

```
-- Fast modular exponentiation by repeated squaring
def modPow (base exp modulus : Nat) : Nat :=
  if modulus ≤ 1 then 0
  else go base exp modulus 1
where
  go (base exp modulus acc : Nat) : Nat :=
    if exp = 0 then acc % modulus
    else
      let base' := (base * base) % modulus
      let acc'  := if exp % 2 = 1 then (acc * base) % modulus else acc
      go base' (exp / 2) modulus acc'
  termination_by exp
```

1.5 Homomorphisms — Structure-Preserving Maps

Definition 1.2 — Group Homomorphism

Let $(G, \cdot_G)$ and $(H, \cdot_H)$ be groups. A function $\varphi : G \to H$ is a **homomorphism** if

$$\varphi(a \cdot_G b) = \varphi(a) \cdot_H \varphi(b) \qquad \text{for all } a, b \in G.$$

If φ is additionally a bijection, it is an **isomorphism**, and we write $G \cong H$.

Lean 4

```
structure GroupHom (G H : Type) [Group G] [Group H] where
  toFun : G → H
  map_mul : ∀ a b : G, toFun (a * b) = toFun a * toFun b

-- A homomorphism preserves the identity
theorem GroupHom.map_one [Group G] [Group H] (φ : GroupHom G H) :
    φ.toFun 1 = 1 := by
  have h := φ.map_mul 1 1
  simp [mul_one] at h  -- Uses 1 * 1 = 1 on the left
  -- Now h : φ.toFun 1 = φ.toFun 1 * φ.toFun 1
  -- Multiplying both sides by (φ.toFun 1)⁻¹ gives the result
  sorry -- Exercise for the reader

-- A homomorphism preserves inverses
theorem GroupHom.map_inv [Group G] [Group H] (φ : GroupHom G H) (a : G) :
    φ.toFun a⁻¹ = (φ.toFun a)⁻¹ := by
  sorry -- Exercise: use map_mul and map_one
```

Exercise 1.4

Fill in both `sorry`s above: prove that a group homomorphism preserves the identity and preserves inverses. *Hint:* For `map_one`, start with the equation $\varphi(1) = \varphi(1 \cdot 1) = \varphi(1) \cdot \varphi(1)$ and multiply both sides on the right by $\varphi(1)^{-1}$.

Exercise 1.5

Define a group homomorphism $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ by $\varphi(k) = k \bmod n$, and prove `map_mul` in Lean.

Chapter 2

Subgroups, Cosets, and Lagrange's Theorem

2.1 Subgroups

Definition 2.1 — Subgroup

Let G be a group. A subset $H \subseteq G$ is a **subgroup** (written $H \leq G$) if:

(i) **Identity:** $e \in H$.
(ii) **Closure:** If $a, b \in H$, then $a \cdot b \in H$.
(iii) **Inverses:** If $a \in H$, then $a^{-1} \in H$.

Lean 4

```
structure Subgroup (G : Type) [Group G] where
  carrier : Set G
  one_mem : (1 : G) ∈ carrier
  mul_mem : ∀ {a b : G}, a ∈ carrier → b ∈ carrier → a * b ∈ carrier
  inv_mem : ∀ {a : G}, a ∈ carrier → a⁻¹ ∈ carrier

-- Membership notation
instance [Group G] : Membership G (Subgroup G) where
  mem a H := a ∈ H.carrier
```

Examples.

-
-
-
-

2.2 Cosets and Lagrange's Theorem

Definition 2.2 — Coset

Let $H \leq G$ and $a \in G$. The **left coset** of H by a is

$$aH = \{a \cdot h : h \in H\}.$$

Theorem 2.1 — Lagrange's Theorem

Let G be a finite group and $H \leq G$. Then $|H|$ divides $|G|$. Moreover, the number of distinct left cosets of H in G is $[G : H] = |G|/|H|$.

Proof.

Lagrange and the "Theorem"

Lagrange never stated this theorem in the language of groups—the concept didn't exist yet. What he showed, in his 1770–1771 *Réflexions sur la résolution algébrique des équations*, was that certain expressions in the roots of a polynomial take on a number of distinct values that divides $n!$ (where n is the degree). This is Lagrange's theorem applied to the symmetric group S_n. The abstract formulation had to wait for Galois, Cayley, and their successors.

2.2.1 Corollaries

Corollary 2.1 — Order of an Element Divides Group Order

If G is a finite group and $g \in G$, then the order of g (the smallest positive n such that $g^n = e$) divides $|G|$.

Proof.

Corollary 2.2 — Euler's Theorem

If $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof. 2.1

Corollary 2.3 — Fermat's Little Theorem

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof.

Euler Proves Fermat

Fermat stated his "little theorem" in 1640 (in a letter to Frénicle de Bessy) without proof. Euler gave the first published proof in 1736, and later generalized it to the theorem bearing his own name. The fact that Fermat's and Euler's theorems are both one-line corollaries of Lagrange's theorem illustrates the power of the group-theoretic framework: results that required individual ingenuity become routine consequences of general structure.

2.3 Normal Subgroups and Quotient Groups

Definition 2.3 — Normal Subgroup

A subgroup $N \leq G$ is **normal** (written $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$, i.e., conjugation by any element of G maps N to itself.
Equivalently: $aN = Na$ for all $a \in G$ (left cosets equal right cosets).

Lean 4

```
structure NormalSubgroup (G : Type) [Group G] extends Subgroup G where
  conj_mem : ∀ {n : G}, n ∈ carrier → ∀ g : G,
    g * n * g⁻¹ ∈ carrier
```

Definition 2.4 — Quotient Group

For $N \trianglelefteq G$, the **quotient group** G/N is the set of left cosets of N, with multiplication $(aN)(bN) = (ab)N$.

Theorem 2.2 — First Isomorphism Theorem

If $\varphi : G \to H$ is a group homomorphism, then:

(i) $\ker(\varphi) = \{g \in G : \varphi(g) = e_H\}$ is a normal subgroup of G.

(ii) $\operatorname{im}(\varphi) = \{\varphi(g) : g \in G\}$ is a subgroup of H.

(iii) $G/\ker(\varphi) \cong \operatorname{im}(\varphi)$.

Exercise 2.1

Compute all subgroups of S_3 and determine which are normal.
Answer (verify in Lean): S_3 has 6 subgroups: $\{e\}$, $\langle s\rangle$, $\langle sr\rangle$, $\langle sr^2\rangle$, $\langle r\rangle = \{e, r, r^2\}$, and S_3 itself. Of these, only $\{e\}$, $\langle r\rangle$, and S_3 are normal.

Exercise 2.2

Prove in Lean that $\ker(\varphi)$ is a normal subgroup for any group homomorphism φ.

Chapter 3

The Subgroup Lattice — Order Theory Interlude

3.1 Why Order Theory?

Dedekind and the Birth of Lattice Theory

Richard Dedekind (1831–1916) was one of the most important algebraists of the 19th century. His contributions include the modern definition of ideals in rings, the Dedekind cut construction of the real numbers, and—most relevant for us—the recognition (around 1897) that the subgroups of a group, ordered by inclusion, form a structure with well-defined "meet" and "join" operations. He called these structures *Dualgruppen*; the term "lattice" was introduced later by Garrett Birkhoff in the 1930s.

Dedekind was also a close collaborator of Cantor and one of the first to recognize the importance of set-theoretic thinking in algebra. His influence on the development of abstract algebra is hard to overstate—Emmy Noether, who revolutionized algebra in the 1920s, explicitly credited Dedekind as her primary inspiration.

3.2 Partial Orders

Definition 3.1 — Partial Order

A **partial order** on a set P is a binary relation $\leq$ that is:

(i) **Reflexive:** $a \leq a$ for all $a \in P$.

(ii) **Antisymmetric:** If $a \leq b$ and $b \leq a$, then $a = b$.

(iii) **Transitive:** If $a \leq b$ and $b \leq c$, then $a \leq c$.

The pair $(P, \leq)$ is called a **partially ordered set** (**poset**).

Lean 4

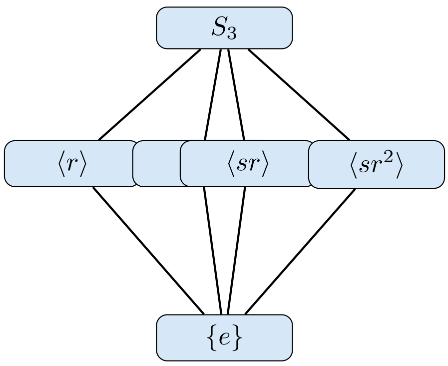
```
class PartialOrder (P : Type) where
  le : P → P → Prop
  le_refl : ∀ a : P, le a a
  le_antisymm : ∀ a b : P, le a b → le b a → a = b
  le_trans : ∀ a b c : P, le a b → le b c → le a c

instance [PartialOrder P] : LE P where le := PartialOrder.le
```

Examples.

-
-
-
-

3.2.1 Hasse Diagrams



Subgroup lattice of S_3

3.3 Lattices

Definition 3.2 — Lattice

A **lattice** is a poset $(L, \leq)$ in which every pair of elements $a, b \in L$ has:

- A **greatest lower bound** (meet, infimum): $a \sqcap b$, satisfying $a \sqcap b \leq a$, $a \sqcap b \leq b$, and if $c \leq a$ and $c \leq b$ then $c \leq a \sqcap b$.
- A **least upper bound** (join, supremum): $a \sqcup b$, satisfying $a \leq a \sqcup b$, $b \leq a \sqcup b$, and if $a \leq c$ and $b \leq c$ then $a \sqcup b \leq c$.

Lean 4

```
class Lattice (L : Type) extends PartialOrder L where
  meet : L → L → L
  join : L → L → L
  meet_le_left  : ∀ a b : L, le (meet a b) a
  meet_le_right : ∀ a b : L, le (meet a b) b
  le_meet : ∀ a b c : L, le c a → le c b → le c (meet a b)
  le_join_left  : ∀ a b : L, le a (join a b)
  le_join_right : ∀ a b : L, le b (join a b)
  join_le : ∀ a b c : L, le a c → le b c → le (join a b) c

instance [Lattice L] : Inf L where inf := Lattice.meet
instance [Lattice L] : Sup L where sup := Lattice.join
```

3.3.1 The Subgroup Lattice

Theorem 3.1 — Subgroups Form a Complete Lattice

For any group G, the set of subgroups of G, ordered by inclusion, forms a complete lattice:

- $H_1 \sqcap H_2 = H_1 \cap H_2$ (intersection of subgroups is a subgroup).
- $H_1 \sqcup H_2 = \langle H_1 \cup H_2 \rangle$ (subgroup generated by the union).

Proof.

Warning

The join of two subgroups is *not* their union. The union of two subgroups is almost never a subgroup (try $\{0, 2, 4\}$ and $\{0, 3\}$ in $\mathbb{Z}/6\mathbb{Z}$: their union contains 2 and 3 but not $2 + 3 = 5$). The join is the *smallest subgroup containing* the union.

3.4 Galois Connections

Definition 3.3 — Galois Connection

Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be posets. A **Galois connection** between P and Q is a pair of maps $\ell : P \to Q$ and $u : Q \to P$ such that:

$$\ell(p) \leq_Q q \iff p \leq_P u(q) \qquad \text{for all } p \in P,\ q \in Q.$$

We call ℓ the **lower adjoint** and u the **upper adjoint**.

Lean 4

```
structure GaloisConnection
    (P Q : Type) [PartialOrder P] [PartialOrder Q] where
  l : P → Q
  u : Q → P
  adjoint : ∀ (p : P) (q : Q), l p ≤ q ↔ p ≤ u q
```

Ore's Abstraction

The concept of a Galois connection was abstracted by Oystein Ore in 1944, who noticed that the specific correspondence Galois discovered between fields and groups (Chapter 8) is an instance of a general pattern appearing throughout mathematics. Ore's paper "Galois Connexions" extracted the essential order-theoretic structure and showed it applies far beyond algebra. The name is fitting: the general concept is named after the specific example that inspired it.

3.4.1 Properties of Galois Connections

Proposition 3.1

Let (ℓ, u) be a Galois connection between P and Q. Then:

(i) Both ℓ and u are **order-reversing**: if $p_1 \leq p_2$ then $\ell(p_2) \leq \ell(p_1)$, and similarly for u.

(ii) **Unit:** $p \leq u(\ell(p))$ for all $p \in P$.

(iii) **Counit:** $\ell(u(q)) \leq q$ for all $q \in Q$.

(iv) **Idempotence:** $\ell \circ u \circ \ell = \ell$ and $u \circ \ell \circ u = u$.

Proof.

Warning

There are two conventions in the literature: some authors define Galois connections with *both maps order-preserving* (the "monotone" variant), and others with *both maps order-reversing* (the "antitone" variant). The Galois correspondence in algebra uses the antitone variant: bigger fields $\leftrightarrow$ smaller groups. We follow the antitone convention.

3.4.2 Closure Operators

Definition 3.4 — Closure Operator

A **closure operator** on a poset $(P, \leq)$ is a map $\mathrm{cl} : P \to P$ that is:

(i) **Extensive:** $p \leq \mathrm{cl}(p)$.
(ii) **Monotone:** $p \leq q \implies \mathrm{cl}(p) \leq \mathrm{cl}(q)$.
(iii) **Idempotent:** $\mathrm{cl}(\mathrm{cl}(p)) = \mathrm{cl}(p)$.

Exercise 3.1

Prove in Lean that $u \circ \ell$ is a closure operator for any (antitone) Galois connection (ℓ, u).

Exercise 3.2

Define the lattice of divisors of 12, ordered by divisibility. Draw its Hasse diagram. Verify that it is a lattice by computing gcd (meet) and lcm (join) for all pairs.

Exercise 3.3

Implement `Lattice (Subgroup G)` in Lean for a finite group `G`. The meet is intersection; the join is the subgroup generated by the union. Prove the lattice axioms.

Chapter 4

Rings and Fields — Adding Multiplication

4.1 The Rise of Abstract Algebra

Hilbert, Noether, and the Abstraction Revolution

David Hilbert (1862–1943) and Emmy Noether (1882–1935) transformed algebra from a collection of computational techniques into a structural science.

Hilbert's 1888 proof of the basis theorem—that every ideal in a polynomial ring is finitely generated—was purely existential, providing no algorithm for finding the generators. Paul Gordan reportedly declared: "This is not mathematics; this is theology!" The controversy between Hilbert's abstract style and Gordan's computational approach foreshadowed the constructivism debate that continues today.

Emmy Noether took the abstraction further. Her 1921 paper *Idealtheorie in Ringbereichen* created modern commutative algebra, showing that the right way to understand a ring is through its *ideals*. As a woman in early 20th-century Germany, she was initially denied a paid position at Göttingen; Hilbert championed her, declaring to the faculty senate: "I do not see that the sex of the candidate is an argument against her admission. After all, we are a university, not a bathing establishment."

She was dismissed in 1933 when the Nazis expelled all Jewish faculty, and emigrated to the United States, where she died at 53. Einstein wrote to the New York Times that she was "the most significant creative mathematical genius thus far produced since the higher education of women began."

4.2 Ring Axioms

Definition 4.1 — Ring

A **ring** $(R, +, \cdot)$ satisfies:
(i) $(R, +)$ is an abelian group with identity 0.
(ii) Multiplication is associative with identity 1.
(iii) **Distributivity:** $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.
If $ab = ba$ for all a, b, the ring is **commutative**.

Lean 4

```
class Ring (R : Type) where
  add : R → R → R
  zero : R
  neg : R → R
  mul : R → R → R
  one : R
  add_assoc : ∀ a b c : R, add (add a b) c = add a (add b c)
  add_comm  : ∀ a b : R, add a b = add b a
  zero_add  : ∀ a : R, add zero a = a
  neg_add   : ∀ a : R, add (neg a) a = zero
  mul_assoc : ∀ a b c : R, mul (mul a b) c = mul a (mul b c)
  one_mul   : ∀ a : R, mul one a = a
  mul_one   : ∀ a : R, mul a one = a
  left_distrib  : ∀ a b c : R, mul a (add b c) = add (mul a b) (mul a c)
  right_distrib : ∀ a b c : R, mul (add a b) c = add (mul a c) (mul b c)
```

4.3 Fields — Where Division Works

Definition 4.2 — Field

A **field** is a commutative ring in which every nonzero element has a multiplicative inverse: for all $a \neq 0$, there exists a^{-1} with $a \cdot a^{-1} = 1$.

Lean 4

```
class Field (F : Type) extends Ring F where
  mul_comm : ∀ a b : F, mul a b = mul b a
  inv : F → F
  mul_inv : ∀ a : F, a ≠ zero → mul a (inv a) = one
  zero_ne_one : zero ≠ one
```

Key examples.

Theorem 4.1 — $\mathbb{Z}/p\mathbb{Z}$ is a Field

For any prime p, $\mathbb{Z}/p\mathbb{Z}$ is a field.

Proof.

4.4 Ideals and Quotient Rings

Definition 4.3 — Ideal

An **ideal** $I \subseteq R$ is an additive subgroup with the absorption property: $r \cdot a \in I$ and $a \cdot r \in I$ for all $r \in R$, $a \in I$.

Theorem 4.2

R/I is a field if and only if I is a **maximal ideal**.

Exercise 4.1

Implement `Field (Fin 5)` in Lean. Compute all multiplicative inverses in $\mathbb{Z}/5\mathbb{Z}$.

Exercise 4.2

Show $\mathbb{Z}/6\mathbb{Z}$ is not a field: find a, b with $a \neq 0$, $b \neq 0$, $ab = 0$.

Chapter 5

Polynomials Over a Field

5.1 The 250-Year Quest

The Hunt for the Quintic Formula

The Babylonians solved quadratics around 2000 BCE. The cubic formula was found around 1515–1535 and published (amid accusations of oath-breaking) by Cardano in 1545. Ferrari found the quartic formula shortly after. Then: 250 years of silence. Euler, Lagrange, Bézout, and many others attempted the quintic, all unsuccessfully. Lagrange (1770) analyzed why the lower-degree methods worked and concluded, presciently, that the same approach couldn't work for degree 5—but he couldn't prove impossibility. The resolution came from Abel (1824) and Galois (1832), whose story we tell in Chapter 10.

5.2 The Polynomial Ring $F[x]$

Definition 5.1 — Polynomial Ring

$F[x]$ is the set of formal expressions $f(x) = a_nx^n + \cdots + a_1x + a_0$ with $a_i \in F$. The **degree** is $\deg(f) = n$ (where $a_n \neq 0$).

Lean 4

```
-- Polynomials as coefficient lists (least-significant first)
structure Polynomial (F : Type) [Field F] where
  coeffs : List F
  deriving Repr

namespace Polynomial

def eval [Field F] (p : Polynomial F) (x : F) : F :=
  p.coeffs.foldr (fun c acc => Ring.add (Ring.mul acc x) c) Ring.zero

def degree [Field F] (p : Polynomial F) : Nat :=
  if p.coeffs.length = 0 then 0 else p.coeffs.length - 1

end Polynomial
```

5.3 Division and Irreducibility

Theorem 5.1 — Division Algorithm

For $f, g \in F[x]$ with $g \neq 0$, there exist unique $q, r \in F[x]$ such that $f = gq + r$ and $\deg(r) < \deg(g)$.

Definition 5.2 — Irreducible Polynomial

$f \in F[x]$ of degree ≥ 1 is **irreducible** over F if it cannot be factored as $f = gh$ with $\deg(g), \deg(h) \geq 1$.

Warning

Irreducibility depends on the field! $x^2 + 1$ is irreducible over $\mathbb{R}$ but factors as $(x+i)(x-i)$ over $\mathbb{C}$, and as $(x+1)^2$ over $\mathbb{Z}/2\mathbb{Z}$.

Theorem 5.2 — Unique Factorization

Every nonzero $f \in F[x]$ factors uniquely (up to order and units) into irreducible polynomials.

Exercise 5.1

Implement polynomial long division in Lean.

Exercise 5.2

Prove that $x^3 - 2$ is irreducible over $\mathbb{Q}$ using the rational root theorem.

Chapter 6

Field Extensions

6.1 Extending the Number System

Kronecker vs. Dedekind

Leopold Kronecker (1823–1891) insisted that new objects must be *constructed* explicitly—a proto-constructivist. His approach to field extensions was concrete: to adjoin a root of an irreducible polynomial $p(x)$, work in $F[x]/(p(x))$. Dedekind preferred an axiomatic approach, defining fields by their properties and proving existence abstractly. Both give the same answer; in Lean, we naturally follow Kronecker.

6.2 The Basic Construction

Definition 6.1 — Simple Algebraic Extension

For irreducible $p \in F[x]$ of degree n:

$$F(\alpha) = F[x]/(p(x)) \cong \{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} : a_i \in F\}$$

where α is the image of x, satisfying $p(\alpha) = 0$.

Lean 4

```
-- ℚ(√2): elements are a + b√2 with a, b ∈ ℚ
structure QSqrt2 where
  a : Rat
  b : Rat
  deriving Repr, DecidableEq

namespace QSqrt2

def add (x y : QSqrt2) : QSqrt2 := ⟨x.a + y.a, x.b + y.b⟩

def mul (x y : QSqrt2) : QSqrt2 :=
  ⟨x.a * y.a + 2 * x.b * y.b, x.a * y.b + y.a * x.b⟩

-- (a + b√2)⁻¹ = (a - b√2)/(a² - 2b²)
def inv (x : QSqrt2) : QSqrt2 :=
  let d := x.a * x.a - 2 * x.b * x.b
  ⟨x.a / d, -x.b / d⟩

def sqrt2 : QSqrt2 := ⟨0, 1⟩

-- Verify: √2 · √2 = 2
example : mul sqrt2 sqrt2 = ⟨2, 0⟩ := by native_decide

end QSqrt2
```

6.3 Degree of an Extension

Definition 6.2 — Degree

$[E : F]$ is the dimension of E as an F-vector space.

Theorem 6.1 — Tower Law

For $F \subseteq K \subseteq L$: $[L : F] = [L : K] \cdot [K : F]$.

6.4 Splitting Fields

Definition 6.3 — Splitting Field

The **splitting field** of $f \in F[x]$ is the smallest extension of F in which f factors into linear factors.

Theorem 6.2 — Existence and Uniqueness

Every nonconstant polynomial has a splitting field, unique up to isomorphism.

The Fundamental Theorem of Algebra

Every polynomial over $\mathbb{C}$ splits completely—$\mathbb{C}$ is algebraically closed. Gauss gave the first rigorous proof in his 1799 doctoral dissertation and returned to this theorem throughout his life, giving four different proofs. The fact that the "fundamental theorem of algebra" requires analysis (not just algebra) for its proof is one of the great ironies of mathematics.

Exercise 6.1

Construct $\mathbb{Q}(\sqrt[3]{2})$ in Lean: elements $a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2$. Implement arithmetic using $(\sqrt[3]{2})^3 = 2$.

Exercise 6.2

Show $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$ using the tower law.

Chapter 7

Field Automorphisms and the Galois Group

7.1 Galois's Story

Évariste Galois (1811–1832)

The life of Évariste Galois is the most dramatic story in mathematics. Born in Bourg-la-Reine, near Paris, he was recognized as a mathematical prodigy by his teachers at the Lycée Louis-le-Grand. At 17, he submitted his first paper to the French Academy of Sciences—it was lost by Cauchy. He submitted again to the Academy's prize competition; the paper was sent to Fourier for review, but Fourier died before reading it, and the manuscript was never found. He was rejected twice from the École Polytechnique (the premier mathematics school in France), apparently because his answers were too far ahead of his examiners.

Meanwhile, Paris was in political turmoil. Galois was a passionate republican, arrested twice for political activities. During his second imprisonment in 1832, he fell in love with Stéphanie-Félicie Poterin du Motel—the details are murky, but it appears a duel resulted, possibly engineered by political enemies. The night before the duel, on May 29, 1832, Galois wrote a long letter to his friend Auguste Chevalier, frantically summarizing his mathematical discoveries. He scrawled "I have no time" in the margins.

The next morning, Galois was shot in the abdomen. He died the following day, at the age of twenty. His brother Alfred gathered his manuscripts. Liouville finally published them in 1846, fourteen years later, recognizing their revolutionary importance.

What Galois created in those few feverish years was nothing less than a new way of thinking about algebra: instead of asking "can I solve this equation?" he asked "what are the *symmetries* of this equation's solutions?" The symmetry group—now called the Galois group—controls everything.

7.2 Field Automorphisms

Definition 7.1 — Field Automorphism

Let E/F be a field extension. A **field automorphism** of E fixing F is a bijection $\sigma : E \to E$ such that:

(i) $\sigma(a + b) = \sigma(a) + \sigma(b)$ for all $a, b \in E$.

(ii) $\sigma(a \cdot b) = \sigma(a) \cdot \sigma(b)$ for all $a, b \in E$.

(iii) $\sigma(a) = a$ for all $a \in F$.

Lean 4

```
structure FieldAut (E F : Type) [Field E] [Field F] where
  toFun  : E → E
  invFun : E → E
  left_inv  : ∀ x, invFun (toFun x) = x
  right_inv : ∀ x, toFun (invFun x) = x
  map_add : ∀ a b, toFun (a + b) = toFun a + toFun b
  map_mul : ∀ a b, toFun (a * b) = toFun a * toFun b
  fixes_base : ∀ a : F, toFun (embed a) = embed a
  -- where embed : F → E is the inclusion map
```

Definition 7.2 — Galois Group

The **Galois group** of E/F, written $\mathrm{Gal}(E/F)$, is the set of all field automorphisms of E fixing F, with composition as the group operation.

7.3 First Example: $\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$

Lean 4

```
-- The nontrivial automorphism of ℚ(√2)
def conjugate : QSqrt2 → QSqrt2
  | ⟨a, b⟩ => ⟨a, -b⟩

-- Verify: conjugate preserves multiplication
theorem conjugate_mul (x y : QSqrt2) :
    conjugate (QSqrt2.mul x y) = QSqrt2.mul (conjugate x) (conjugate y) := by
  simp [conjugate, QSqrt2.mul]
  ring  -- or explicit computation

-- Verify: conjugate is an involution (applying it twice gives identity)
theorem conjugate_conjugate (x : QSqrt2) :
    conjugate (conjugate x) = x := by
  simp [conjugate]; ring

-- Verify: conjugate fixes ℚ (elements with b = 0)
theorem conjugate_fixes_rat (a : Rat) :
    conjugate ⟨a, 0⟩ = ⟨a, 0⟩ := by
  simp [conjugate]; ring
```

7.4 Normal and Separable Extensions

Definition 7.3 — Normal Extension

E/F is **normal** if every irreducible polynomial in $F[x]$ that has at least one root in E splits completely in E.

Definition 7.4 — Separable Extension

E/F is **separable** if every element of E is a root of a *separable* polynomial over F (one with no repeated roots).

Definition 7.5 — Galois Extension

E/F is a **Galois extension** if it is both normal and separable. Equivalently, $|\operatorname{Gal}(E/F)| = [E : F]$.

Dedekind and Artin

Galois himself didn't separate the concepts of normality and separability—the distinction was clarified by Dedekind in the 1870s and given its modern form by Emil Artin in the 1940s. Artin's reformulation of Galois theory, replacing Galois's original (somewhat obscure) arguments with clean linear algebra, is the version we follow in this book.

Intuition

Normal ensures there are *enough* automorphisms. Separable ensures they're all *distinct*. Together: $|\operatorname{Gal}(E/F)| = [E : F]$, so the Galois group is "as large as it could possibly be."

Exercise 7.1

Compute $\mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$. Show it is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (the Klein four-group) by exhibiting all four automorphisms.

Exercise 7.2

Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* a normal extension. *Hint:* $x^3 - 2$ has one root in $\mathbb{Q}(\sqrt[3]{2})$ but not all three (the other two are complex).

Chapter 8

The Fundamental Theorem of Galois Theory

8.1 The Main Event

Theorem 8.1 — Fundamental Theorem of Galois Theory

Let E/F be a finite Galois extension with Galois group $G = \mathrm{Gal}(E/F)$. There is an order-reversing bijection

$$\{\text{intermediate fields } K : F \subseteq K \subseteq E\} \underset{\text{Inv}}{\overset{\text{Fix}}{\rightleftarrows}} \{\text{subgroups } H \leq G\}$$

given by:

$$\mathrm{Fix}(K) = \mathrm{Gal}(E/K) = \{\sigma \in G : \sigma(x) = x \text{ for all } x \in K\}$$
$$\mathrm{Inv}(H) = E^H = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in H\}$$

Moreover:

(i) Fix and Inv are mutual inverses: $\mathrm{Inv}(\mathrm{Fix}(K)) = K$ and $\mathrm{Fix}(\mathrm{Inv}(H)) = H$.

(ii) The correspondence **reverses order**: $K_1 \subseteq K_2$ iff $\mathrm{Fix}(K_2) \leq \mathrm{Fix}(K_1)$.

(iii) **Degrees match indices:** $[E : K] = |\,\mathrm{Fix}(K)|$ and $[K : F] = [G : \mathrm{Fix}(K)]$.

(iv) K/F is a normal (Galois) extension iff $\mathrm{Fix}(K) \trianglelefteq G$. In that case, $\mathrm{Gal}(K/F) \cong G/\,\mathrm{Fix}(K)$.

Intuition

The fundamental theorem says: **the structure of a field extension is entirely determined by the structure of its symmetry group.** Bigger intermediate fields have *fewer* symmetries (more elements are pinned down), so the corresponding subgroup is smaller. This is an *antitone Galois connection*—exactly the concept from Chapter 3! In fact, this *is* the original Galois connection. The general concept (Chapter 3) was abstracted from this specific example by Ore in 1944.

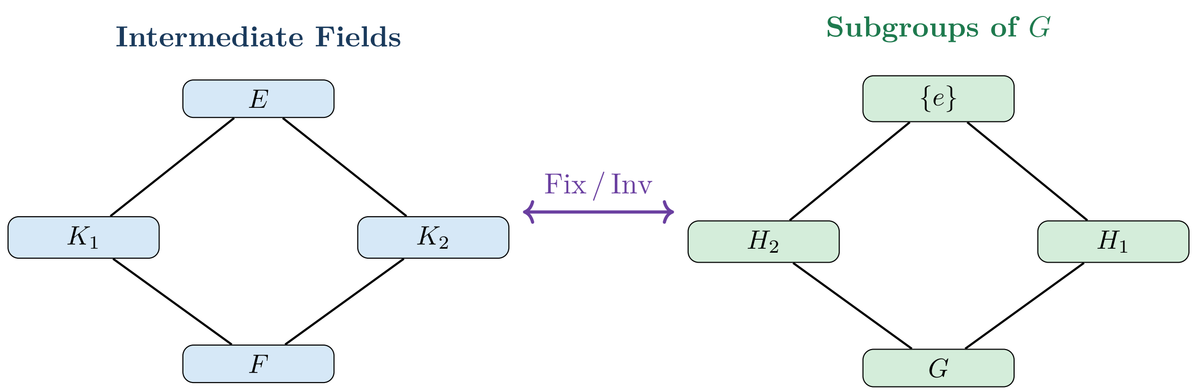
Lean 4

```
-- The Fixed Field map: subgroup H ↦ elements fixed by H
def fixedField (H : Subgroup (Gal E F)) : IntermediateField E F where
  carrier := { x : E | ∀ σ ∈ H.carrier, σ.toFun x = x }
  -- proof obligations: show this is a subfield
  sorry

-- The Fixing Group map: intermediate field K ↦ automorphisms fixing K
def fixingGroup (K : IntermediateField E F) : Subgroup (Gal E F) where
  carrier := { σ : Gal E F | ∀ x ∈ K.carrier, σ.toFun x = x }
  -- proof obligations: show this is a subgroup
  sorry

-- The Galois connection: these maps form an antitone Galois connection
-- between the lattice of intermediate fields and the lattice of subgroups
theorem galois_connection :
    ∀ K H, fixingGroup K ≤ H ↔ Inv H ≤ K := by
  sorry  -- The core of the fundamental theorem
```

8.2 The Two Lattices



Exercise 8.1

Verify the fundamental theorem for $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$: list all intermediate fields, list all subgroups of $\mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and check the correspondence.

Chapter 9

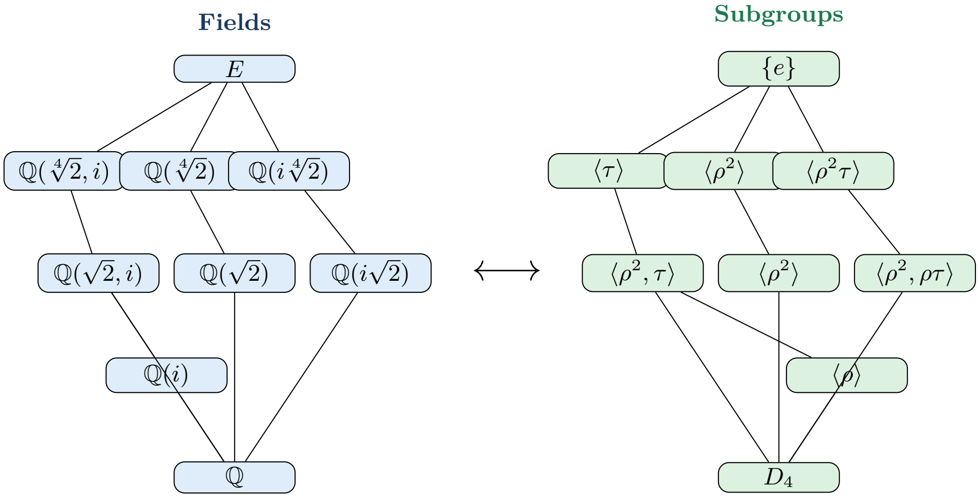
Worked Example — The Splitting Field of $x^4 - 2$

9.1 Building the Splitting Field

9.2 Computing the Galois Group

-
-

9.3 The Correspondence



Exercise 9.1

Verify the correspondence for at least three intermediate field–subgroup pairs: show that the automorphisms fixing the field form the claimed subgroup.

Chapter 10

Solvability — Why There Is No Quintic Formula

10.1 Two Tragic Heroes

Abel and Galois

Niels Henrik Abel (1802–1829) grew up in poverty in Norway. At 22, he proved that the general quintic equation cannot be solved by radicals—the first impossibility result in algebra. He sent his proof to Gauss (who ignored it) and to Cauchy at the French Academy (who lost the manuscript). Abel spent two years traveling Europe trying to gain recognition, returned to Norway in debt, contracted tuberculosis, and died at 26. Two days later, a letter arrived offering him a prestigious professorship in Berlin.

Galois (Chapter 7) went further: he found the exact criterion for when a polynomial *is* solvable by radicals. The criterion is stated in terms of the Galois group.

Both men died tragically young—Abel at 26, Galois at 20. Their work was recognized only posthumously. Today, abelian groups bear Abel's name, and Galois theory is the cornerstone of modern algebra.

10.2 Solvable Groups

Definition 10.1 — Solvable Group

A group G is **solvable** if there exists a chain of subgroups

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_n = G$$

where each quotient G_{i+1}/G_i is abelian.

Lean 4

```
-- A composition series: chain of normal subgroups with abelian quotients
def IsSolvable (G : Type) [Group G] : Prop :=
  ∃ (n : Nat) (chain : Fin (n + 1) → Subgroup G),
    chain ⟨0, by omega⟩ = ⊥ ∧                -- starts at {e}
    chain ⟨n, by omega⟩ = ⊤ ∧                -- ends at G
    ∀ i : Fin n,                              -- each step is normal with
      IsNormal (chain i) (chain i.succ) ∧     -- abelian quotient
      IsAbelian (Quotient (chain i.succ) (chain i))
```

10.3 The Solvability Criterion

Theorem 10.1 — Galois's Criterion

A polynomial $f \in F[x]$ is solvable by radicals if and only if its Galois group $\mathrm{Gal}(E/F)$ (where E is the splitting field of f) is a solvable group.

10.4 S_n for $n \leq 4$ Is Solvable

Proposition 10.1

S_4 is solvable, with composition series:

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4$$

where $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$ is the Klein four-group and A_4 is the alternating group. The quotients are: $V_4/\{e\} \cong V_4$ (abelian), $A_4/V_4 \cong \mathbb{Z}/3\mathbb{Z}$ (abelian), $S_4/A_4 \cong \mathbb{Z}/2\mathbb{Z}$ (abelian).

10.5 S_5 Is Not Solvable

Theorem 10.2 — A_5 Is Simple

The alternating group A_5 (order 60) has no nontrivial normal subgroups.

Proof sketch.

Corollary 10.1

S_5 is not solvable.

Proof.

Theorem 10.3 — The General Quintic Is Unsolvable

There is no formula in radicals for the roots of a general degree-5 polynomial.

Proof.

The Punchline

The jump from S_4 to S_5 is where simplicity (in the technical sense) first appears. A_4 has the Klein four-group as a nontrivial normal subgroup; A_5 has none. There is something fundamentally more complicated about permuting 5 objects than permuting 4. Galois theory makes this precise: the additional complexity is exactly what prevents a radical formula from existing.

Exercise 10.1

Prove in Lean that S_3 is solvable by exhibiting the composition series $\{e\} \trianglelefteq \langle r \rangle \trianglelefteq S_3$.

Exercise 10.2

Prove that every group of order ≤ 4 is solvable. *Hint:* every group of prime order is cyclic, hence abelian, hence solvable.

Chapter 11

Finite Fields — The Complete Classification

11.1 Galois's Legacy

The Fields That Bear His Name

In his final manuscripts, Galois described the finite fields: for every prime power p^n, there is exactly one field of that size (up to isomorphism), and these are *all* the finite fields. They are denoted $\mathrm{GF}(p^n)$ ("Galois field") or $\mathbb{F}_{p^n}$ in his honor.

Theorem 11.1 — Classification of Finite Fields

(i) For every prime p and positive integer n, there exists a field of order p^n.
(ii) Any two fields of order p^n are isomorphic.
(iii) Every finite field has order p^n for some prime p and some $n \geq 1$.

11.2 Constructing Finite Fields

Definition 11.1

$\mathrm{GF}(p^n) = \mathbb{F}_p[x]/(f(x))$ where f is any irreducible polynomial of degree n over $\mathbb{F}_p$.

-
-
-

Exercise 11.1

Find an irreducible polynomial of degree 2 over $\mathbb{F}_3$ and use it to construct GF(9). Write out the addition and multiplication tables.

Chapter 12

$GF(2^8)$ and the AES S-Box — Capstone

12.1 AES and Galois Fields

The Advanced Encryption Standard

The Advanced Encryption Standard (AES), adopted by NIST in 2001 after a five-year international competition, encrypts the vast majority of the world's internet traffic. Its designers—Joan Daemen and Vincent Rijmen (Rijndael)—chose to build the cipher's core nonlinear operation around arithmetic in $GF(2^8)$ because:

- Addition in $GF(2^8)$ is XOR—free in hardware.
- Multiplication can use lookup tables—fast in software.
- The multiplicative inverse has excellent cryptographic properties.

12.2 Constructing $GF(2^8)$

Lean 4

```
-- GF(2⁸): elements are bytes, representing polynomials over F₂
structure GF256 where
  val : UInt8
  deriving DecidableEq, Repr

namespace GF256

-- Addition = XOR (polynomial addition mod 2)
def add (a b : GF256) : GF256 := ⟨a.val ^^^ b.val⟩

-- Multiplication = polynomial multiplication mod m(x)
-- m(x) = x⁸ + x⁴ + x³ + x + 1 = 0x11B
def mul (a b : GF256) : GF256 :=
  -- Russian peasant multiplication with reduction
  go a.val b.val 0
where
  go (a b acc : UInt8) : GF256 :=
    if b == 0 then ⟨acc⟩
    else
      let acc' := if b &&& 1 != 0 then acc ^^^ a else acc
      let a'   := if a &&& 0x80 != 0
                  then (a <<< 1) ^^^ 0x1B  -- reduce mod m(x)
                  else a <<< 1
      go a' (b >>> 1) acc'
  termination_by b.toNat

-- Multiplicative inverse via extended Euclidean algorithm
-- (or: a⁻¹ = a^(254) since a^(255) = 1 in GF(2⁸)×)
def inv (a : GF256) : GF256 :=
  -- a^254 = a^(2+4+8+16+32+64+128)
  -- Compute by repeated squaring
  let a2   := mul a a
  let a4   := mul a2 a2
  let a8   := mul a4 a4
  let a16  := mul a8 a8
  let a32  := mul a16 a16
  let a64  := mul a32 a32
  let a128 := mul a64 a64
  mul (mul (mul (mul (mul (mul a2 a4) a8) a16) a32) a64) a128

-- The AES S-Box: inverse then affine transform
def sbox (a : GF256) : GF256 :=
  let b := if a.val == 0 then ⟨(0 : UInt8)⟩ else inv a
  affineTransform b
where
  affineTransform (b : GF256) : GF256 :=
    -- The affine transformation specified by AES
    -- b' = Ab + c where A is a specific 8×8 matrix over F₂
    -- and c = 0x63
    sorry -- Matrix multiplication over F₂

end GF256
```

12.3 Proving Properties

-
-
-

Exercise 12.1

Prove that GF(2^8) satisfies the field axioms. The hardest part is showing associativity of multiplication (polynomial multiplication mod $m(x)$). For a finite type, `native_decide` can handle this.

Exercise 12.2

Compute the full AES S-box table (all 256 entries) and verify it against the published AES specification.

Chapter 13

Compass and Straightedge — Impossibility Proofs

13.1 Three Ancient Problems

2000 Years of Frustration

Three construction problems haunted mathematicians from ancient Greece until the 19th century:

1. **Trisecting an angle:** Given an arbitrary angle, divide it into three equal parts.
2. **Doubling the cube:** Given a cube, construct another with exactly twice the volume.
3. **Squaring the circle:** Given a circle, construct a square with the same area.

All three were to be done with compass and straightedge alone. Generations of mathematicians attempted these constructions and failed. The Galois-theoretic framework explains *why*.

13.2 Constructible Numbers

Theorem 13.1 — Constructibility Criterion

A real number α is constructible if and only if there exists a tower of field extensions

$$\mathbb{Q} = F_0 \subset F_1 \subset \cdots \subset F_n \ni \alpha$$

with $[F_{i+1} : F_i] = 2$ for each i. In particular, $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ must be a power of 2.

13.3 Trisecting 60°

13.4 Doubling the Cube

13.5 Squaring the Circle

Gauss and the 17-gon

On the positive side: Gauss proved at age 19 (1796) that the regular 17-gon is constructible. He was so proud of this that he reportedly asked for a regular 17-gon to be inscribed on his tombstone. The criterion: a regular n-gon is constructible iff $n = 2^k \cdot p_1 \cdots p_m$ where the p_i are distinct Fermat primes ($p = 2^{2^j} + 1$). The known Fermat primes are 3, 5, 17, 257, and 65537.

Exercise 13.1

Prove in Lean that $8t^3 - 6t - 1$ has no rational roots by checking all possibilities from the rational root theorem.

Chapter 14

Error-Correcting Codes and Reed-Solomon

14.1 Codes Over Finite Fields

Reed and Solomon (1960)

Irving Reed and Gustave Solomon, at MIT Lincoln Laboratory, described a class of error-correcting codes based on polynomial evaluation over finite fields. Reed-Solomon codes protect data on CDs, DVDs, Blu-rays, QR codes, deep-space probes (Voyager), RAID storage, and digital television. The mathematics is pure Galois theory.

Exercise 14.1

Implement a simple Reed-Solomon encoder over $GF(2^8)$: given a message of k bytes, produce n evaluation points.

Chapter 15

Galois Connections Revisited

15.1 The Big Picture

Theorem 15.1

The following are all instances of Galois connections:

(i) **Galois theory:** intermediate fields $\leftrightarrow$ subgroups.

(ii) **Algebraic geometry (Nullstellensatz):** affine varieties $\leftrightarrow$ radical ideals. $V(I)$ = common zeros of I; $I(V)$ = polynomials vanishing on V.

(iii) **Topology:** interior $\leftrightarrow$ closure. Open sets form a lattice; closed sets form a lattice; interior and closure connect them.

(iv) **Logic:** theories $\leftrightarrow$ models. $\mathrm{Th}(M)$ = sentences true in models M; $\mathrm{Mod}(T)$ = models satisfying theory T.

(v) **Computer science:** abstract interpretation. Concrete semantics $\leftrightarrow$ abstract domains. The abstraction and concretization maps form a Galois connection.

Intuition

The pattern: whenever you have two "worlds" connected by maps that translate between them, and the translations approximately invert each other (with a controlled loss of information), you likely have a Galois connection. The "closed" elements on each side—those not affected by the round trip—form the part of each world that is perfectly captured by the other.

Exercise 15.1

Verify that the maps V and I from algebraic geometry satisfy the Galois connection axioms (for a polynomial ring over an algebraically closed field).

Chapter 16

Looking Back — The Unity of Algebra

16.1 The Story Arc

16.2 What We Proved

-
-
-
-

16.3 The Galois-Theoretic Worldview

Hilbert's 12th Problem and Beyond

In 1900, David Hilbert posed 23 problems for the 20th century. His 12th problem asked for an explicit construction of all abelian extensions of arbitrary number fields—a generalization of Galois theory that remains open in full generality. The Langlands program, one of the deepest ongoing research efforts in mathematics, can be seen as a vast generalization of Galois theory connecting number theory, representation theory, and geometry. Hilbert would have been amazed.

16.4 Connections Forward

"The algebraic symmetries of the roots of an equation contain the secret of its solvability."

— The message of Galois theory,
written in haste the night of May 29, 1832.

Appendix: Lean 4 Typeclass Hierarchy